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$$I_m = 2.2 \times \sqrt{2} = 3.1A$$

Therefore,

$$\upsilon = 311 \sin 314t$$

$$i = 3.1 \sin\left(314t + \pi/2\right)$$

4.7.6 AC circuit containing a resistor, an inductor and a capacitor in series – Series RLC circuit

Consider a circuit containing a resistor of resistance R, a inductor of inductance L and a capacitor of capacitance C connected across an alternating voltage source (Figure 4.51). The applied alternating voltage is given by the equation.

$$\upsilon = V_m \sin \omega t \qquad (4.52)$$

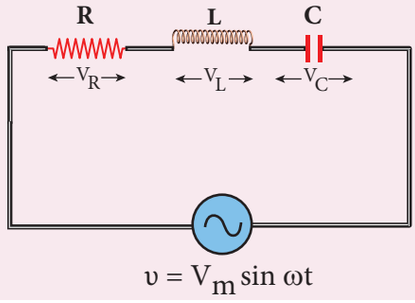


Figure 4.51 AC circuit containing R, L and C

Let i be the resulting circuit current in the circuit at that instant. As a result, the voltage is developed across R, L and C.

We know that voltage across R (V_R) is in phase with i, voltage across L (V_L) leads i by $\pi/2$ and voltage across C (V_C) lags i by $\pi/2$.

The phasor diagram is drawn with current as the reference phasor. The current is represented by the phasor $\overrightarrow{OI}$, V_R by $\overrightarrow{OA}$; V_L by $\overrightarrow{OB}$ and V_C by $\overrightarrow{OC}$ as shown in Figure 4.52.

The length of these phasors are

$$OI = I_m,\ OA = I_m R,\ OB = I_m X_L;\ OC = I_m X_C$$

The circuit is either effectively inductive or capacitive or resistive that depends on the value of V_L or V_C. Let us assume that $V_L > V_C$ so that net voltage drop across L-C combination is $V_L - V_C$ which is represented by a phasor $\overrightarrow{AD}$.

By parallelogram law, the diagonal $\overrightarrow{OE}$ gives the resultant voltage υ of V_R and $(V_L - V_C)$ and its length OE is equal to V_m. Therefore,

$$V_m^2 = V_R^2 + \left(V_L - V_C\right)^2$$

$$= \sqrt{\left(I_m R\right)^2 + \left(I_m X_L - I_m X_C\right)^2}$$

$$= I_m \sqrt{R^2 + \left(X_L - X_C\right)^2}$$

$$\text{or} \quad I_m = \frac{V_m}{\sqrt{R^2 + \left(X_L - X_C\right)^2}} \qquad (4.53)$$

$$\text{or} \quad I_m = \frac{V_m}{Z}$$

$$\text{where} \quad Z = \sqrt{R^2 + \left(X_L - X_C\right)^2} \qquad (4.54)$$

Z is called impedance of the circuit which refers to the effective opposition to the circuit current by the series RLC circuit. The voltage triangle and impedance triangle are given in the Figure 4.53.

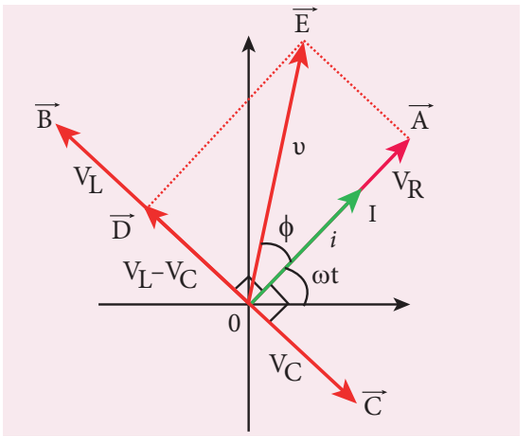


Figure 4.52 Phasor diagram for a series RLC – circuit when $V_L > V_C$

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